

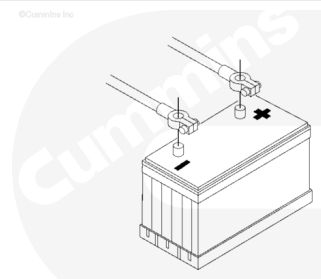
## Trainings ▾

### Preparatory Steps

#### Front Gear Train

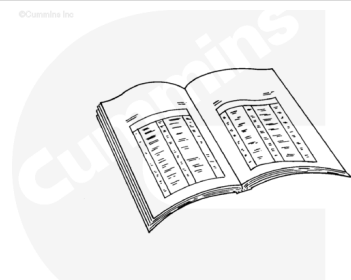
##### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



- Disconnect the batteries.

- Disconnect the cold starting aid (if equipped). Refer to the OEM manual
- Remove the air crossover tube (if equipped). Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Disconnect the charged air cooler hose (if equipped). Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html>)
- Remove the high-pressure fuel lines. Refer to Procedure 006-051 (</qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html>)
- Remove the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>).



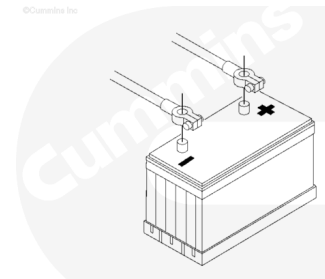
#### Rear Gear Train

##### **⚠ WARNING ⚠**

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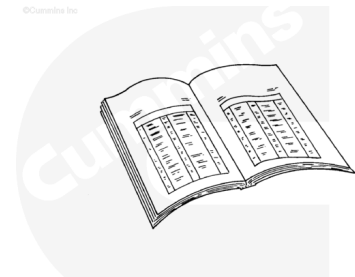
**remove the negative (-) battery cable first and attach the negative (-) battery cable last.**

- Disconnect the batteries.



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- Remove the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>)
- Remove the closed crankcase ventilation hose. Refer to Procedure 003-024 (</qs3/pubsys2/xml/en/procedures/40/40-003-024-tr.html>)
- Disconnect the cold starting aid. Refer to the OEM manual
- Remove the air intake connection adapter. Refer to Procedure 010-131 (</qs3/pubsys2/xml/en/procedures/40/40-010-131-tr.html>)
- If required, remove the air intake heater element. Refer to Procedure 010-124 (</qs3/pubsys2/xml/en/procedures/40/40-010-124-tr.html>)
- Remove the dipstick tube. Refer to Procedure 007-011 (</qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html>)
- Remove the fuel supply line. Refer to Procedure 006-024 (</qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html>)
- Remove the fuel drain line. Refer to Procedure 006-021 (</qs3/pubsys2/xml/en/procedures/40/40-006-021-tr.html>)
- Remove the high pressure fuel lines. Refer to Procedure 006-051 (</qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html>).

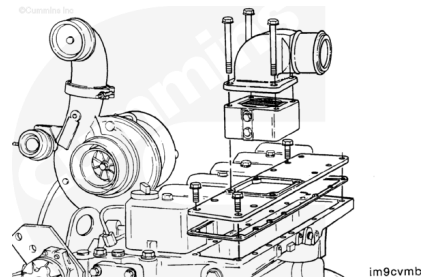


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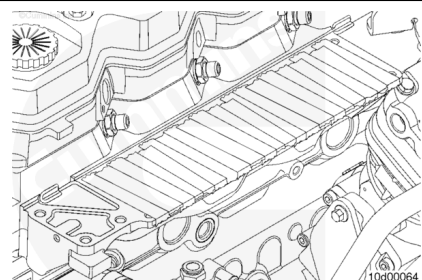
## Remove

### Front Gear Train

Remove the manifold cover, gasket, and grid heater (if equipped).



If the air intake manifold will be off for a prolonged period of time, tape off the intake manifold opening to prevent debris from entering the intake system.

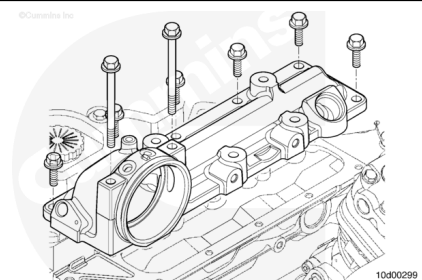


## Rear Gear Train

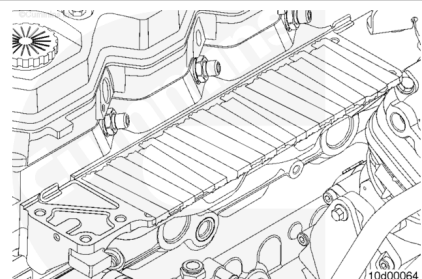
Remove the intake manifold mounting capscrews.

**Note :** The intake manifold uses RTV sealant between the manifold and cylinder head. Be careful not to damage the intake manifold and sealing surfaces when removing the intake manifold.

Remove the intake manifold.



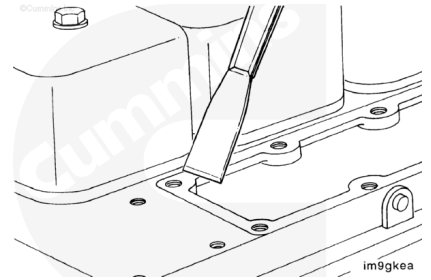
If the air intake manifold will be off for a prolonged period of time, tape off the intake manifold opening to prevent debris from entering the intake system.



## Clean and Inspect for Reuse

**⚠ WARNING ⚠**

**When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.**

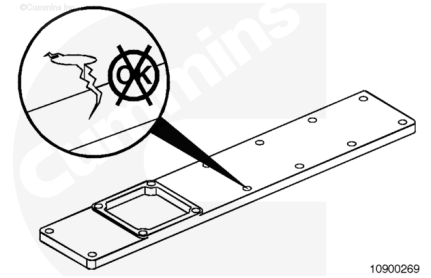


Clean the air intake manifold and cylinder head sealing surfaces with a gasket scraper and solvent.

**Note :** Keep the gasket material and any other material out of the air intake.

Check the air intake manifold for damage.

Replace the air intake manifold, if necessary.

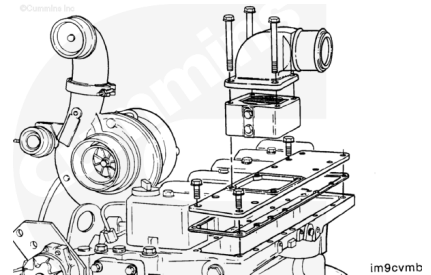


## Install

### Front Gear Train

Install the cover and a new gasket.

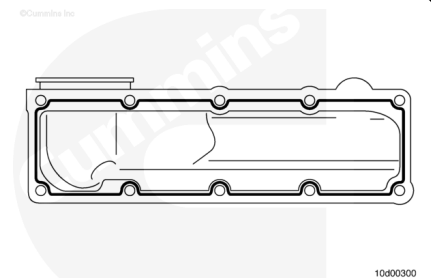
**Torque Value:** 24 n•m [ 18 ft•lb ]



### Rear Gear Train

Apply RTV sealant, Part Number 3164070, to the intake manifold in the path illustrated.

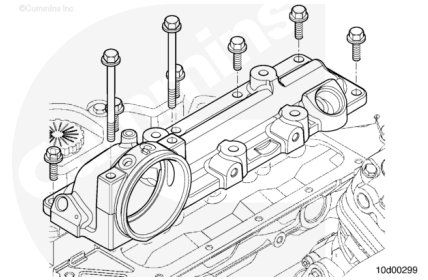
**Note :** Install the gear housing within 10 minutes of applying sealant or the sealant will not seal correctly. Once installed, allow the to dry for 30 minutes before running the engine.



Install the intake manifold and mounting capscrews.

Tighten the mounting capscrews.

**Torque Value:** 24 n•m [ 18 ft-lb ]



## Finishing Steps

### Front Gear Train

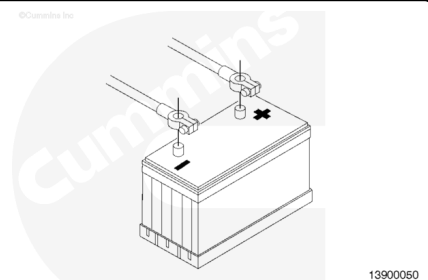
- Install the high-pressure fuel lines. Refer to Procedure 006-051 (</qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html>)
- Install the air intake piping. Refer to Procedure 010-024 (</qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html>)
- Install the air crossover tube (if removed). Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html>)
- Install the charged air cooler hose (if removed). Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html>)
- Connect the cold starting aid (if removed). Refer to the OEM manual.



### **⚠ WARNING ⚠**

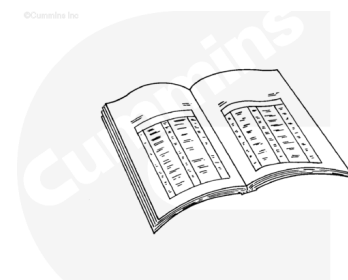
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- Connect the batteries.



### Rear Gear Train

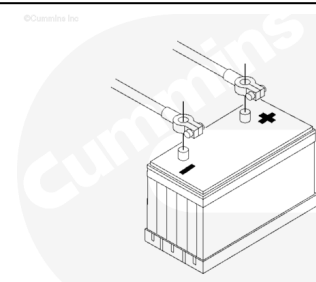
- Install the high pressure fuel lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/40/40-006-051-tr.html)
- Install the fuel supply line. Refer to Procedure 006-024 (/qs3/pubsys2/xml/en/procedures/40/40-006-024-tr.html)
- Install the fuel drain line. Refer to Procedure 006-021 (/qs3/pubsys2/xml/en/procedures/40/40-006-021-tr.html)
- Install the dipstick tube. Refer to Procedure 007-011 (/qs3/pubsys2/xml/en/procedures/40/40-007-011-tr.html)
- If removed, install the air intake heater element. Refer to Procedure 010-124 (/qs3/pubsys2/xml/en/procedures/40/40-010-124-tr.html)
- Install the air intake connection adapter. Refer to Procedure 010-131 (/qs3/pubsys2/xml/en/procedures/40/40-010-131-tr.html)
- Install the closed crankcase ventilation hose. Refer to Procedure 003-024 (/qs3/pubsys2/xml/en/procedures/40/40-003-024-tr.html)
- Connect the cold starting aid. Refer to the OEM manual
- Install the air intake piping. Refer to Procedure 010-024 (/qs3/pubsys2/xml/en/procedures/40/40-010-024-tr.html)



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- Connect the batteries
- Operate engine and check for leaks.

